

REPORT ON SAMPLE SALES DATA

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Data Overview

The Samples Sales database is a dataset containing about 3,000 order records from 2003–2005. It consists of 7 core tables: Customers, Orders, OrderDetails, Products, ProductLines, Employees, and Offices.

The USA, France, and Australia generate the highest revenue, while Motorcycles and Classic Cars are the top-selling product lines.

The data follows the 80/20 rule, where a small percentage of customers contribute most of the revenue.

It is ideal for SQL practice, joins, sales analysis, ranking, time analysis, and dashboard reporting.

Explanation of the Queries Used

A. Ranking & Top N

- Top 10 by quantity, sales, revenue, orders: Queries 1, 2, 3, 4, 15, 27, 28, 32 use TOP 10 + ORDER BY DESC. These find best/worst performers fast.
- Ranking within groups: Queries 21, 22, 26 use ROW_NUMBER() and RANK() with PARTITION BY. Example: Top products within each product line and Top customers per country.
- Latest record per group: Query 23 uses ROW_NUMBER() OVER (PARTITION BY CustomerID ORDER BY OrderDate DESC) to get the latest order per customer.

B. Aggregations & Grouping

- Revenue and volume by dimension: Queries 6, 7, 11, 12, 29, 30, 31,34,38,39 aggregate SUM(Sales), SUM(QuantityOrdered) by DealSize, ProductLine, Country, State, QTR_ID, Status.
- Customer behavior: Queries 4, 13, 24,35,36,40 aggregate order count and spend per customer. Query 13 uses HAVING COUNT > 5 to filter loyal customers.
- Average calculations: Query 8 computes true average order value by first aggregating to order level in a subquery.

C. Joins

- INNER JOIN: Most queries join Customers-Orders-OrderDetails-Products to combine customer, product, and sales data.
- LEFT JOIN: Query 17 finds all customers including those with zero orders.
- RIGHT JOIN: Query 19 finds orders with missing customer records. Its useful for data quality checks.
- LEFT JOIN + IS NULL: Query 18 finds products never sold.

D. Subqueries & CTEs

- Above average filters: Queries 9, 20, 25 use subqueries and CTEs to compare each customer/product to the average.
- CTEs for readability: Queries 10, 25, 26,33 use CTEs to break multi-step logic like monthly sales → YoY growth.

E. Window Functions

- Running totals: Query 24 uses SUM() OVER (ORDER BY YEAR_ID, MONTH_ID) for cumulative revenue.
- YoY growth: Query 26,37 uses LAG(Revenue, 12) to compare same month last year.
- Partitioned rankings: Query 26 also uses SUM() OVER (PARTITION BY YEAR_ID) for YTD revenue.

Key Index Recommendations

1. Core Join paths

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CREATE INDEX IX_Orders_CustomerID ON Orders(CustomerID);
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```
CREATE INDEX IX_OrderDetails_OrderNumber ON OrderDetails(OrderNumber);
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CREATE INDEX IX_OrderDetails_ProductCode ON OrderDetails(ProductCode);
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2. Time-based analysis

CREATE INDEX IX_Orders_YEAR_MONTH ON Orders(YEAR_ID, MONTH_ID);

CREATE INDEX IX_Orders_QTR_ID ON Orders(QTR_ID);

CREATE INDEX IX_Orders_OrderDate ON Orders(OrderDate);

3. Filtering & grouping

CREATE INDEX IX_Customers_Country ON Customers(Country);

CREATE INDEX IX_Customers_State ON Customers(State);

CREATE INDEX IX_Products_ProductLine ON Products(ProductLine);

CREATE INDEX IX_OrderDetails_DealSize ON OrderDetails(DealSize);

4. Covering index for top product queries

CREATE INDEX IX_OrderDetails_ProductCode_Sales

ON OrderDetails(ProductCode) INCLUDE (Sales, QuantityOrdered);

Insights

- Motorcycles and Classic Cars generate the highest revenue and sales volume.
- A small number of products contribute most of the total revenue, following the Pareto principle.
- Some products show little or no sales activity, indicating slow-moving or inactive inventory.
- High MSRP products tend to generate stronger revenue despite lower order frequency.
- A small percentage of customers generate a large portion of total revenue.
- Repeat customers contribute significantly to business growth.
- High-frequency customers usually have higher lifetime value.
- Customers with above-average spending are ideal targets for retention and loyalty programs.

- USA and France are the strongest revenue-generating regions.
- Q4 consistently performs better than other quarters, showing seasonal demand patterns.
- Large deal sizes produce higher revenue but occur less frequently.
- Monthly trend analysis shows periods of strong and weak sales performance throughout the year.
- Some products have no associated sales records.
- A few orders may exist without matching customer records.
- Missing regional information such as State values can affect geographic reporting accuracy.
- Disputed or cancelled orders should be monitored closely to reduce revenue loss.

Recommendations

- Increase marketing and inventory for top-performing product lines such as Motorcycles and Classic Cars.
- Develop loyalty programs for repeat and high-value customers.
- Focus sales campaigns on high-performing countries and regions.
- Review or discontinue products with no sales activity.
- Maintain adequate inventory for high-demand products to prevent stock shortages.
- Monitor disputed and cancelled orders to improve operational efficiency.
- Segment customers based on spending and purchase frequency.
- Offer personalized promotions to high-value customers.
- Encourage repeat purchases through discounts and loyalty rewards.
- Use monthly and yearly revenue trends for demand forecasting.
- Track KPIs such as:
 - Average order value
 - Revenue growth
 - Repeat customer rate
 - Top customer contribution
- Automate reporting dashboards for faster business decision-making.